

Google: LC 3025 - Find the Number of Ways to Place People I

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Problem Statement

given number of coordinates, we need to find the number of pairs (A, B) such that A is on upper left side of B and no point lie inside the rectangle they make.

Example & Visualization

Points: [(6,2), (4,4), (2,6)]

Note: (2,6) and (6,2) form a rectangle.

This pair is **not valid** as (4, 4) lies in the rectangle made by (2,6) and (6,2).

$\because 2 < 4 < 6$ and $2 < 4 < 6$

So O/P = 2

Bruteforce Approach

```
for (each point A) {  
    for (each point B) {  
        if A and B does not satisfy condition then continue  
  
        bool valid = true;  
  
        for (each point C) {  
            if (C inside region) {  
                valid = false;  
                break;  
            }  
        }  
  
        if (valid) ans++;  
    }  
}
```

TC = $O(n^3)$

SC = $O(1)$

C++ Implementation

```
class Solution {
public:
    int numberOfPairs(vector<vector<int>>& points) {

        int n = points.size();
        int ans = 0;

        for(int i=0; i<n; i++){
            for(int j=0; j<n; j++){

                if(j == i) continue;

                int x1 = points[i][0];
                int x2 = points[j][0];
                int y1 = points[i][1];
                int y2 = points[j][1];

                if(!(x1 <= x2 && y1 >= y2)){
                    continue; //valid pair he hi nhi to further check ka matlab nhi he
                }

                bool flag = true; //maan ke chalo valid he

                for(int k=0; k<n; k++){
                    if(k == i || k == j){
                        continue; //same point nhi consider krna
                    }

                    int x3 = points[k][0];
                    int y3 = points[k][1];

                    if(x1 <= x3 && x3 <= x2 && y2 <= y3 && y3 <= y1){
                        flag = false;
                        break;
                    }
                }

                if(flag) ans++;
            }
        }
        return ans;
    }
};
```